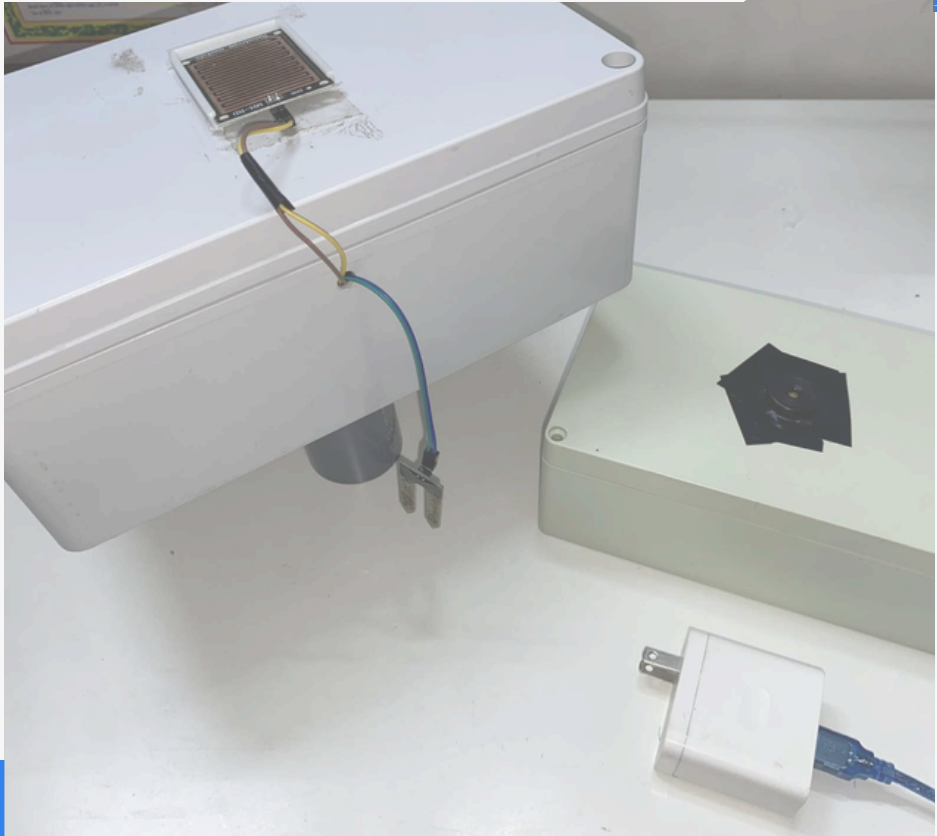


ICIA 2025

Landslide Detection Warning System



Stay Alert. Stay Safe. Stay Ahead.

A smart landslide warning system combining real-time environmental sensors and web monitoring, enabling early alerts to protect lives, infrastructure, and support disaster management decisions.

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“ Discover our smart Landslide Warning System—how it works, its components, key features, and real-world impact. Learn how it helps prevent disasters and empowers communities with timely, life-saving alerts. ”

01 Problems

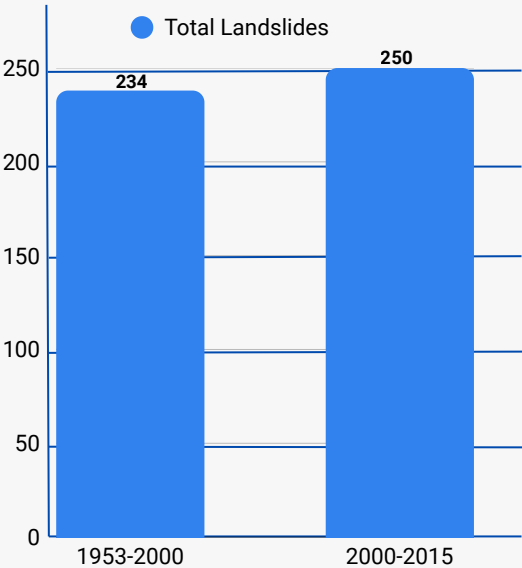
Reality

Landslides are deadly and globally impactful.

From 2004 to 2016, landslides took nearly **56,000 lives** across **4,862 disasters** worldwide.



According to the **Vietnam Ministry of Industry and Rural Development**:



Landslides are becoming more frequent

There were 234 landslides recorded between 1953 - 2000 and 250 landslides between 2000 - 2015. **The frequency of landslides has been increasing over time**, with the average number of landslides per year rising from 4 to 15-16

Statistics from Nu Village (Vietnam, 09/2024) reveal the brutal impact of landslides

1.6M³

Over **1.6 million cubic meters** of mud and rock raced down 3.6 kilometers, burying 37 houses in Nu Village.

52

The death toll has risen to **52**, with 15 injured and 14 missing.

5'

The flash flood swept away the entire village in just **5 minutes**, an astonishingly rapid disaster.



Landslide scene at Nu Village

01 Problems

Paint Points



Cong Vinh (52), resident at Marine Apartment

- A long-distance driver.
- As a long-haul driver, he once faced a sudden landslide on the Hai Van Pass. Though he escaped safely, the fear stuck with him.
- Now, he always stays alert in landslide-prone areas, especially in the rain.
- He wishes for real-time road condition updates.



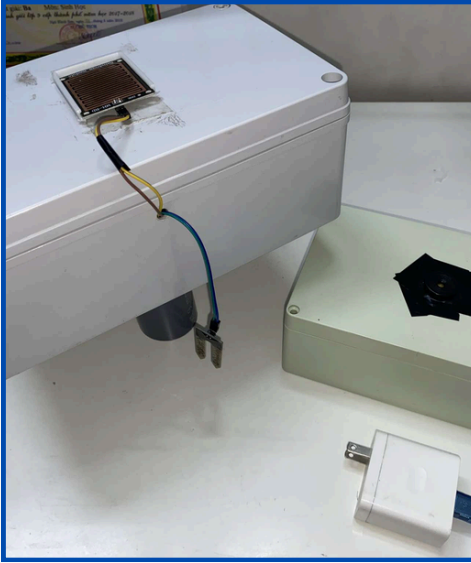
Thanh Hao (46), local resident at the base of Son Tra mountain

- After a small landslide damaged his family's car, he became worried about his personal property.
- He had missed the warnings from local authorities.
- He wishes for timely, direct updates on landslide risks, so he can act early instead of relying solely on official alerts.

Landslides are a constant concern, and users need proactive tools to stay prepared and protected.

01 Problems

Goals



Our invention focus on:

- Landslide prediction by sensors
- Speaker warning in the surrounding area
- Storing information, creating landslide maps



Good health and well-being



Protecting public health through early warnings that reduce casualties and prevent landslide-related injuries.

Sustainable cities and communities



Enhancing disaster-resilient communities by supporting local authorities in landslide prevention and mitigation.

Climate Action

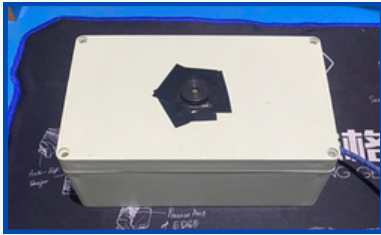


Addressing climate-related disasters by using environmental sensors to detect risks, minimize damage.

Our product consists of two main components: **hardware** and **software**.



Sensor devices



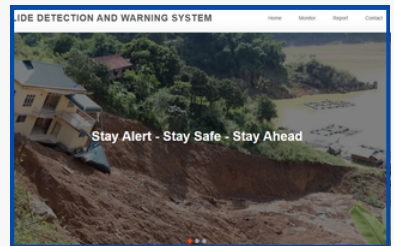
The warning system

Hardware includes:

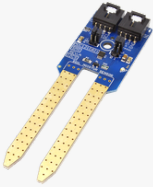
- **Sensor devices** are placed in high-risk or previously affected landslide areas, designed to withstand extreme weather and collect real-time data.
- **The warning system**, installed in populated areas near landslide zones, triggers loudspeaker alerts when the risk level reaches a critical threshold.

Our website offers a powerful and user-friendly interface that allows:

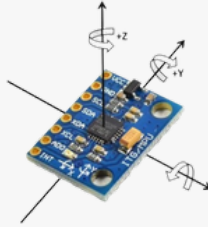
- Monitors sensor status, connectivity, and power in real time.
- Displays an interactive map with device locations, risk levels, and historical data.
- It also provides risk analysis and lets residents report hazards, send feedback, and upload images.



To allow product collect real-time environmental data, the product need some key components:



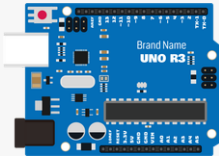
Moisture Sensor



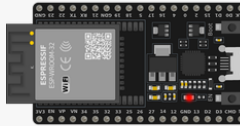
Accelerometer Sensor



Rain Sensor



Arduino Uno R3



ESP32



Alarm Siren



LoRa Transmitter
Module



LoRa Receiver Module



Power Bank

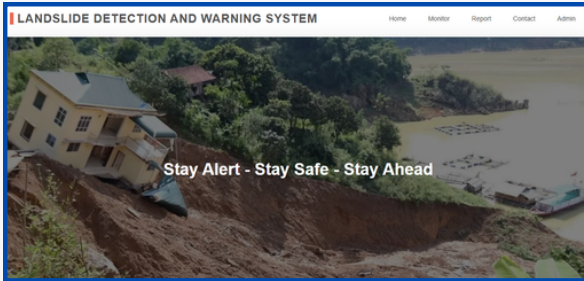


Waterproof Enclosure
150x80x70 (mm)



Waterproof Enclosure
200x150x100 (mm)

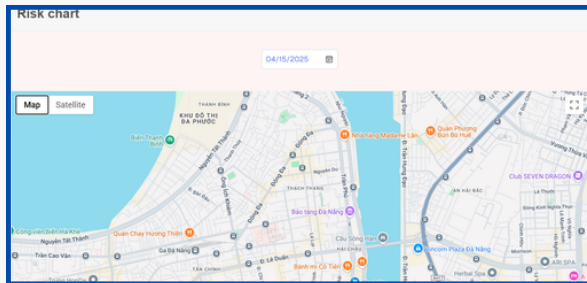
The software brings everything together - connecting data, devices, and people through a user-friendly platform



Home page

The Home Page

Providing an overview of the system, its purpose, and visual context of real landslide disasters.



Landslide map

The Device Map Page

Displaying an interactive map with all sensor placements, helping users identify at-risk areas geographically.

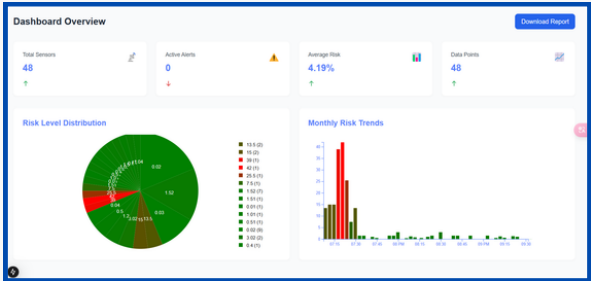


Risk chart

The Risk Chart Page

Visualizing real-time and historical landslide risk levels through intuitive graphs, helping users track changes and trends over time.

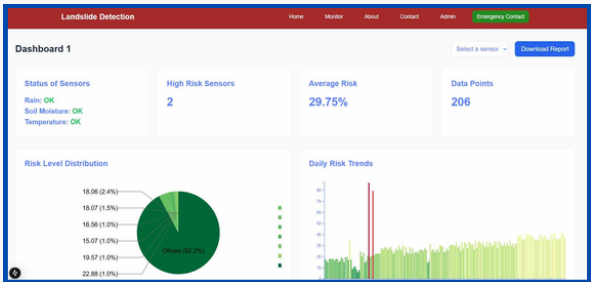
Now, let's explore the management pages - where real-time data, device control, and risk analysis come together to ensure faster, smarter disaster response.



All products management

The All Products Management Page

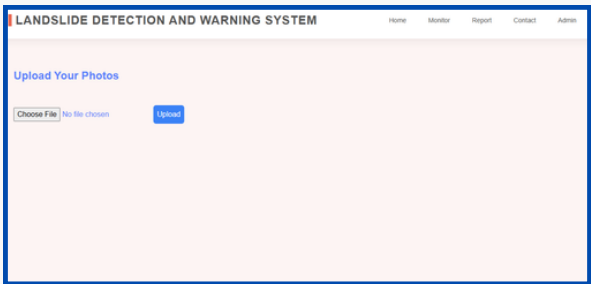
Allowing administrators to monitor every deployed unit, including risk levels and sensor status at a glance.



Single product management

The Single Product Management Page

Showing detailed stats of one device, including daily risk charts, sensor data, and device health.



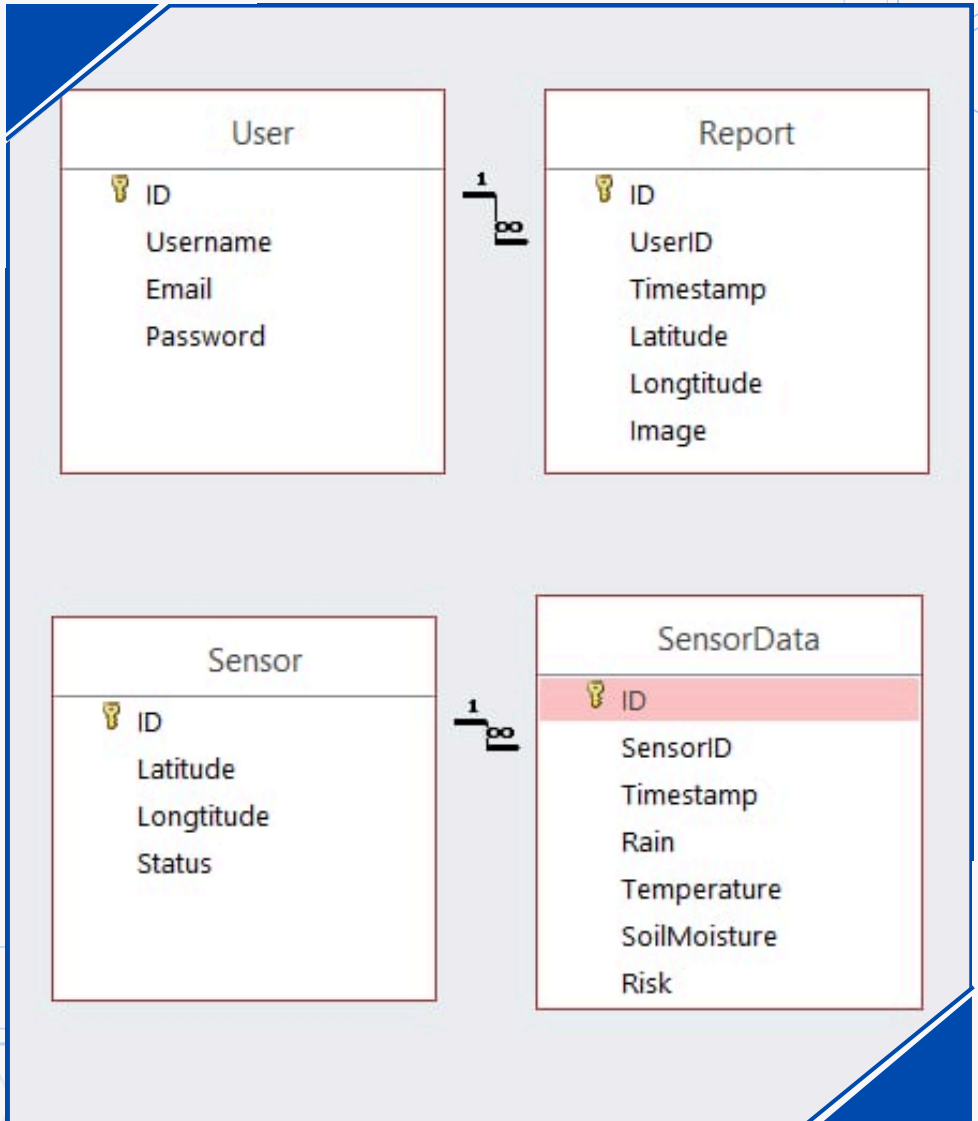
Landslide report

The Report Page

Allows users to submit feedback, images, and information about areas they believe are at risk of landslides.



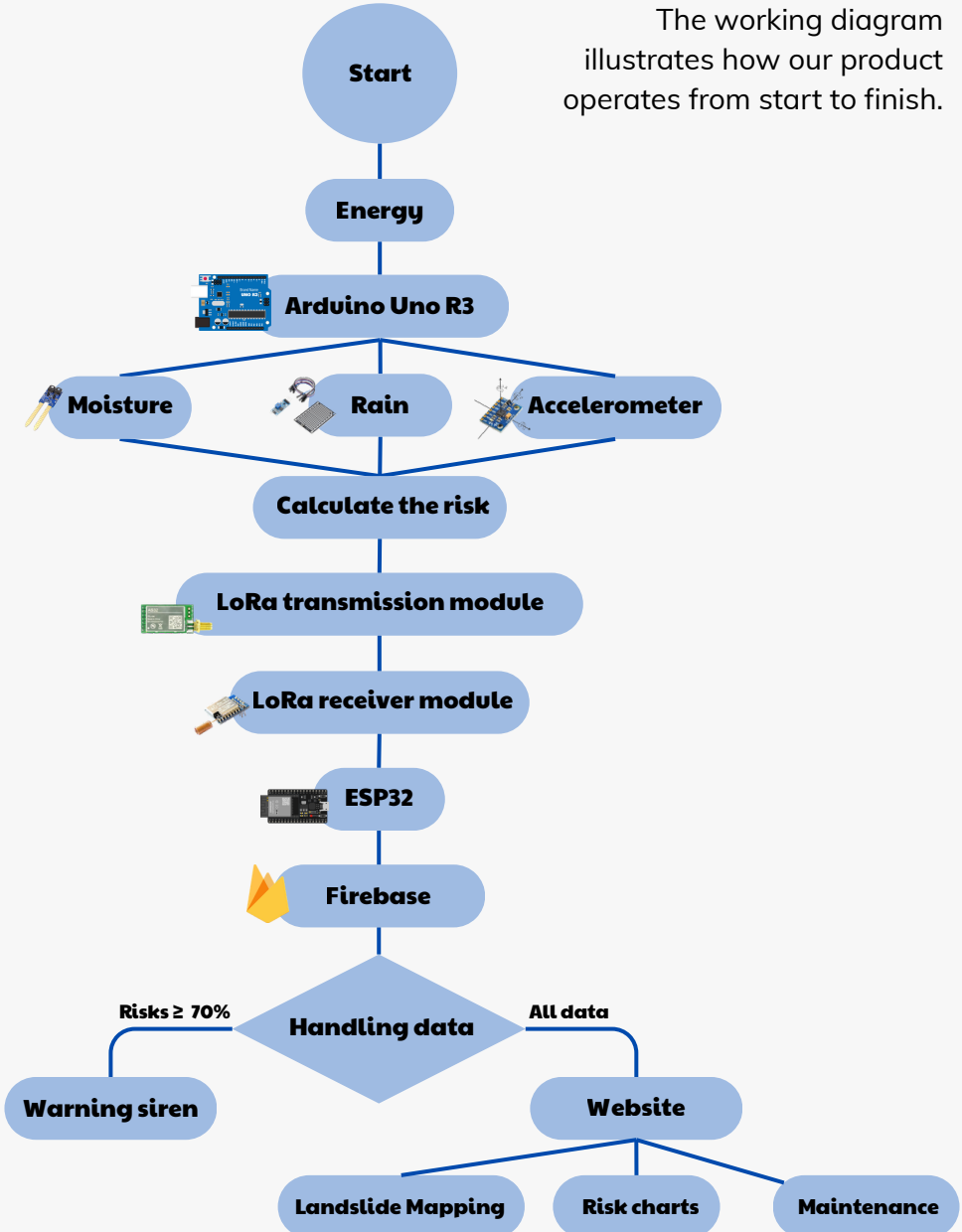
Our system uses **Firebase** to store sensor data, track device status, manage user feedback, and enable real-time updates across the monitoring and warning platform.

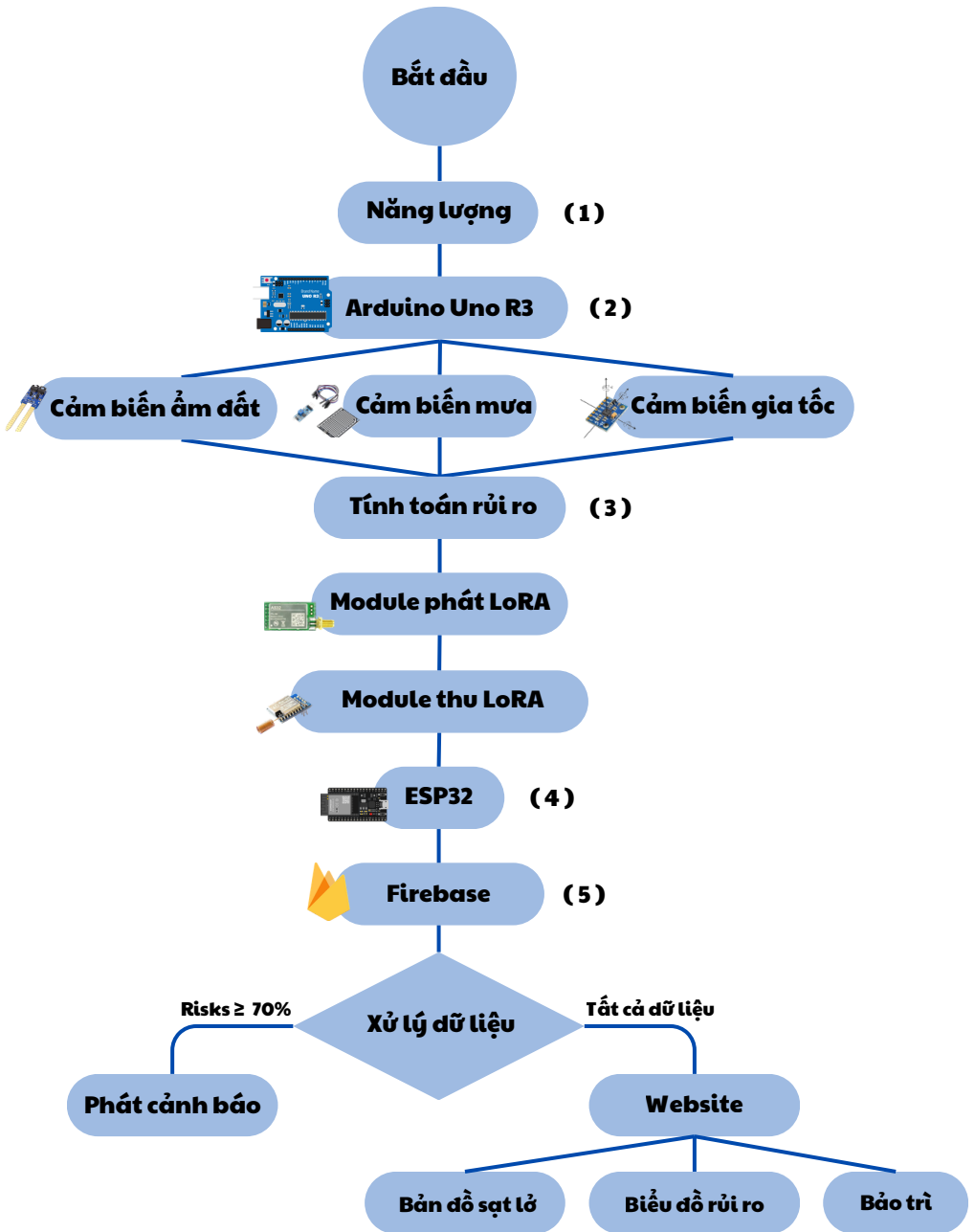


02 Solutions

Operational flowchart

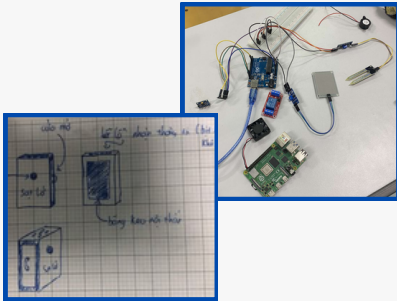
The working diagram illustrates how our product operates from start to finish.





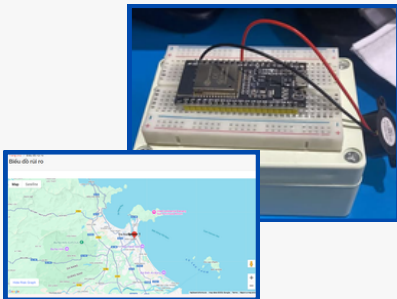
Our current system is Version 3. But to get here, we went through multiple iterations, each one solving a different challenge

Version 1



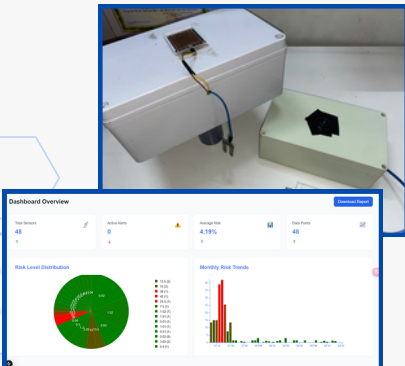
- Components: Sensors, Arduino Uno R3, Raspberry Pi 4
- Function: Detection only - no alerts, no website
- Limitations: Could not withstand harsh environments, no communication system, and high cost due to Raspberry Pi

Version 2



- Improvements: Added protective casing for sensor devices
- Introduced: Basic alert system and web platform (landslide map, risk charts)
- Limitation: Depended on Wi-Fi — limited deployment range in remote areas

Version 3 (Current)



- Upgraded: Replaced Raspberry Pi with ESP32 for cost efficiency
- Added: Protective case for alert devices
- Communication: Switched to LoRa - long-range, low-power transmission
- Web Platform: Expanded with real-time management pages and device monitoring

02 Solutions

Risk formula

The formula for calculating the product risk is calculated using the formula taken from the research article on NHESS.

Shear Strength

$$s = c' + (p - u_w) \cdot \tan \phi'$$

- **s**: Shear strength, resists soil from sliding down
- **c'**: Effective cohesion of soil
- **p**: Total normal stress on the slip surface
- **u**: Pore water pressure
- **ϕ'** : Effective internal friction angle

Surface Runoff

$$Q_c = (I - I_0) \cdot D$$

- **Q_c** : Surface runoff volume
- **I**: Rainfall intensity
- **I_0** : Average infiltration rate
- **D**: Duration of rainfall

The risk formula is calculated based on shear strength and surface runoff.

$$\text{Risk level (R)} = \frac{\text{Shear Strength (s)}}{\text{Surface Runoff (Q}_c\text{)}} \cdot 100$$

Risk levels are classified as follows:

- $R \geq 70\%$: High
- $40\% < R < 70\%$: Medium
- $R \leq 40\%$: Low



03 Results

Early Warning Saves Lives



Real-time alerts allow communities to evacuate and take action before landslides strike, preventing loss of life.

Support for Disaster Management



Delivers crucial data, risk maps, and monitoring tools to help local authorities plan and respond more effectively.

Empowers Local Communities



Provides residents direct risk updates, reducing reliance on central alerts and boosting preparedness.



Thanh Hao (46), local resident at the base of Son Tra mountain

User Feedback

- This system never existed in his area.
- He believe the system would be highly beneficial if widely implemented.
- It could help significantly reduce the impact of natural disasters.

Our team conducted field testing at Son Tra Mountain from January 12 to January 14 and obtained the following results:

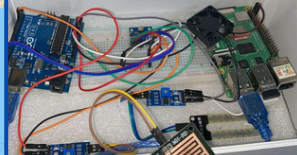


The results show that the system operates effectively under natural environmental conditions:

- Sensors functioned accurately and consistently.
- The signal transmission to the warning device (placed 1km away) was fast and stable.
- The website updated charts and other information in near real-time.
- The power supply needs to be upgraded for more reliable long-term operation.

04Summary

Comparing our product to other products:

Names	Student Research Topic (2011-2015)	Current Vietnam System	Our Product
Products	 		
Sensor	Accelerometer sensor	Based on rainfall	Based on multiple sensors
Detection	Low	High	High
Warning	Send warning to authorities	No direct warning	Direct warning to both people and government
Website	None	Shows landslide risk areas	Provides landslide risk and area, other information

- **Sensor Technology:** Our product uses multiple sensors (moisture, rain, and accelerometer).
- **Warning System:** Our system sends direct warnings to both people and authorities, improving safety and response time
- **Website Integration:** Our product includes a website that provides real-time risk levels, sensor data, landslide maps, and user feedback features

Overall: Our system is more comprehensive, user-centered, and reliable, enhancing both prevention and preparedness for landslides.

Future Development

To ensure the system grows stronger, smarter, and more sustainable, we focus on three key areas of improvement:

Durable Sensor Integration



Upgrade sensors that can deliver more stable and long-lasting performance.

Feature Expansion



Add mobile notifications, satellite data integration, ect to create a complete landslide preparedness platform

Efficient Power System



Developing a low-cost solar solution with accurate battery monitoring

Acknowledgements

We extend our heartfelt thanks to the individuals and institutions who made this project possible:

- **Nguyen Huu Sieu**, for his invaluable guidance and unwavering support throughout our development journey.
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- **Le Duc Chinh, Do Tran Khanh Vinh, and Van Cong Nam**, for their dedication, teamwork, and consistent contributions to the project.

Your support has been instrumental in transforming our vision into reality.



<https://landslide-ledws.vercel.app/>



Access additional
information by QR code.

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